

SHREE VARAA MANGAI V

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EDUCATION

Master of Science in Artificial Intelligence Systems | GPA: 3.90 / 4.0

Aug 2024 - May 2026

University of Florida, Gainesville, FL

Relevant Coursework: Deep Learning in Medical Image Analysis, ML for AI Systems, Applied Deep Learning, AI Ethics, Computational Photography, Computer Vision, NLP

Bachelor of Technology in Information Technology | GPA: 8.43 / 10

Sep 2020 - May 2024

Loyola-ICAM College of Engineering and Technology, India (affiliated under Anna University)

Relevant Coursework: OOP Principles, Computational Intelligence, Computer Architecture, Cryptography & Network Security, Design and Analysis of Algorithms, Big Data Analytics, Linear Algebra, Statistics, Database Management Systems, Data Structures

SKILLS

AI/ML: PyTorch, Tensorflow, VLLM, Hugging Face Transformers, LLMs, RAG, LangChain, LangGraph, LlamaIndex, MCP, CUDA

Backend & Data: Python, FastAPI, PostgreSQL, Supabase/pgvector, Redis, MongoDB, Pinecone, REST APIs

Cloud & Infrastructure: Docker, Linux, Git, CI/CD, AWS Bedrock, Azure, NVIDIA NIM, Prometheus, Grafana

Frontend: TypeScript, JavaScript, React.js, Next.js

WORK AND RESEARCH EXPERIENCE

ML Research | University of Florida- Trustworthy Engineered Autonomy Lab

Florida, US | Sep 2025 - Present

Research lab focused on reliable and interpretable AI systems. This role centred on benchmarking hallucination in text-to-video generative models.

- Developed a multimodal hallucination detection pipeline using **Qwen3-VL-30B**, evaluating **15,000+** Wan 2.1 and HunyuanVideo generations on NVIDIA B200 GPUs across T2VCompBench and ViBe.
- Designed and benchmarked a token-level prompt consistency detector against **four VLM baselines**, achieving the strongest performance across all six evaluation metrics on T2VCompBench with Wan 2.1.

ML Engineering Co-op | University of Florida- IPPD | AGIS Inc.

Florida, US | Aug 2025 - Apr 2026

It's an enterprise AI company that builds custom AI-powered tools for business automation.

- Ran the 2D Gaussian Splatting (2DGS) reconstruction stage end-to-end on **NVIDIA B200 GPU** clusters, using SAM2-based zero-shot object segmentation, COLMAP feature extraction/pose estimation, and Gaussian optimization, achieving **32.72 dB PSNR** / 0.0059 L1 render error.
- Benchmarked end-to-end output against Kiri Engine on identical inputs: delivered a Unity-ready rigged asset in **8 min vs. Kiri's 15 min** for a raw PLY export, a **47% latency reduction** with richer output.
- Profiled GPU workloads on NVIDIA B200 clusters using PyTorch/CUDA tooling, optimizing inference or reconstruction latency through batching, memory-aware preprocessing, and throughput benchmarking.

Global People Analytics Intern | Ford Motor Company

Tamil Nadu, IN | Aug 2023 - Oct 2023

Ford's Global People Analytics team uses data-driven methods to inform HR strategy. This internship focused on workforce forecasting and budget planning.

- Conducted statistical analysis and forecasting on time series forecasting models such as Autoregressive Integrated Moving Average (ARIMA), Vector Autoregression (VAR), and Vector Error Correction Model (VECM) and non-time series models, such as Lasso and Ridge regression, to determine optimal female incumbency.
- Developed ETL pipelines in Python to predict salary costs using workforce datasets containing 1,000+ employee records, improving **model accuracy by 20%** through exploratory data analysis, feature engineering, and data quality improvements.

Software Engineer Intern | Spacescan Ltd.

Ontario, Canada | Sep 2022 - Jan 2023

Spacescan is a blockchain explorer and analytics platform. This internship involved full-stack development of the consumer-facing web and mobile application.

- Built and maintained responsive, component-based React.js interfaces integrating Swagger-documented REST APIs, reducing code duplication through reusable UI components and robust state management.
- Extended the React Native mobile application with new features, collaborating on frontend architecture and software design decisions to maintain consistent UI and interaction patterns across web and mobile.

PROJECTS

CareMind: Agentic Clinical RAG Assistant | [GitHub-Link](#)

Aug 2026-Sept 2026

- Built an agentic RAG system using FastAPI, LangGraph, Supabase/pgvector, Redis, NVIDIA NIM, and MCP, orchestrating document QA, report comparison, and medical education workflows with tool calling and context-aware routing.
- Developed an evaluation and observability pipeline using Recall@K, Precision@K, MRR, faithfulness, latency, and TTFT with Prometheus/Grafana, improving **retrieval recall from 0.598 to 0.764** and **faithfulness support from 0.505 to 0.677**.

Tech Stack: Python, FastAPI, LangGraph, Supabase/pgvector, Redis, NVIDIA NIM, MCP, Prometheus, Grafana, Next.js, TypeScript

Multi-Image Super Resolution for Prostate MRI (CNN, SRGAN, Diffusion Models) | [GitHub-Link](#)

Nov - Dec 2025

- Trained and benchmarked **three architectures**: a 5-block Residual Interpolation CNN, SRGAN (PatchGAN discriminator with adversarial/L1 loss), and a **Fast-DDPM diffusion model** (10-step DDIM, non-uniform β schedule) on 40,532 slice-pairs to reconstruct missing prostate MRI through-plane slices.
- SRGAN led on numerical accuracy (29.045 dB PSNR, 0.850 SSIM vs. CNN's 29.027 dB/0.843), while Fast-DDPM traded **PSNR (27.019 dB, 0.801 SSIM)** for the strongest anatomical continuity, **at 24x the inference latency** (10.5ms vs. 0.4-0.6ms).

Tech Stack: PyTorch, CNN, GAN, Diffusion Models, OpenCV, NumPy

CognitoMap: Automated Question Classification using Bloom's Taxonomy | [GitHub-Link](#)

Feb - Apr 2024

- Built a scalable ETL pipeline extracting and transforming questions from PDFs into MongoDB, achieving **94% accuracy** in question extraction using the Gemini 1.0 Pro model and classified questions into Bloom's Taxonomy levels using Sentence Transformer embeddings with Pinecone vector retrieval.
- Delivered a full-stack dashboard (Flask, React.js, Chart.js) giving educators real-time insights into the cognitive distribution of their assessments across Bloom's Taxonomy levels.

Tech Stack: Flask, React, MongoDB, Pinecone, Chart.js, TypeScript

CERTIFICATIONS

- Building with the Claude API:** Anthropic (2026) · **Build & Orchestrate Agents with MS Foundry:** Microsoft (2026) · **Fundamentals of Deep Learning:** NVIDIA (2024) · **Fine-Tuning LLMs for Cybersecurity:** LinkedIn Learning (2025) · **Tesla Stock Price Prediction / Google Cloud Fundamentals:** Coursera (Dec 2023 / Aug 2022)